

SHELSON SHELLY

Chennai, Tamil Nadu, India | +91 7907780833 | shelsonshelly@gmail.com
linkedin.com/in/shelsonshelly | github.com/shxlson

EDUCATION

SRM Institute of Science & Technology - Kattankulathur

Bachelor of Technology in Computer Science Engineering, Specialization in AI & ML

2023 - 27

CGPA: 8.94

SKILLS

Languages: Python, SQL, C, C++

Software Engineering: Data Structures & Algorithms, OOP, FastAPI, REST APIs, Backend Development, Microservices, Data Pipelines, Debugging & Troubleshooting

Cloud & DevOps: AWS, Azure, Docker, Jenkins (CI/CD), Git, Bitbucket, Version Control

AI/ML & Data Skills: CNN, ANN, Machine Learning, Deep Learning, Computer Vision, LLMs, Data Analysis, Explainable AI

Process & Delivery: Stakeholder Management, Agile & Scrum, Project Planning, Code Reviews, Technical Documentation, Root-Cause Analysis, Technical Presentation & Communication

EXPERIENCE

Research Intern, Tata Steel — TMT Steel Analyzer

Jan 2026 - Jun 2026

Center of Excellence for Agentic Twins

- Designed and engineered a scalable, modular **computer vision pipeline (Python)** achieving **96.4% overall classification accuracy** on a dataset of **500+ test samples**, extracting 6 geometric features (rib count, spacing, angle, height, length, AR value) across rib and ring tests.
- Developed precision measurement **algorithms** and rule-based decision logic, reducing **Mean Absolute Error (MAE) to 0.118 mm** for diameter calculations and **1.06° for transverse angles**, enabling reliable, standardized accept/reject decisions.
- Optimized image-processing routines and system performance, achieving an average **inference time of 1.32 seconds** per image while maintaining **>96% Precision and Recall (F1 Score: 96.6%)** under lighting variability, surface defects, and motion blur.

Research Intern, Mahindra Research Valley

Oct 2025 - Mar 2026

- Built a **full-stack computer vision system (Python, OpenCV)** processing real-time feeds from **industrial camera streams**, achieving **93.1% event-detection accuracy** and enabling automated, data-driven sustainability workflows across plant operations.
- Architected scalable, low-latency video-processing and spatial-analysis **pipelines** with edge inference, reducing average per-frame decision latency to **<190 ms** and cutting manual monitoring effort by an estimated **30%**.

PROJECTS

Interpretable Deep-fake Detection using VXAI

- Designed an interpretable deep-fake detection **framework** achieving **93.2% classification accuracy** on benchmark datasets, combining deep learning-based synthetic media detection with feature-level visual explanations (XAI), under UROP - SRMIST KTR (Advisor: Asst. Prof. Dr. Kiruthika M.).
- Improved model interpretability and reliability, reducing the **false-positive rate to 4.8%** and strengthening trustworthiness of AI-driven decisions for media forensics and misinformation detection.

DiagramSync AI - Hackathon Project (MeetStream AI, OpenAI, Python)

- Built an **AI agent** that auto-generates UML, ER, and flow diagrams from Zoom/Google Meet transcripts, processing **150+ meeting transcripts** at **89% diagram-generation accuracy** using an NLP pipeline.
- Integrated MeetStream AI and OpenAI **APIs** to automate technical documentation, cutting manual diagramming time by an estimated **65%** and placing **Top 10 of 100+ teams**.

ACHIEVEMENTS

1st Runner-up - AWS Generative AI Ideathon

Amazon Web Services (AWS) - AWS PartyRock

- Secured **2nd position** among participating teams in a Generative AI ideathon organized by AWS for selected students.
- Built an AI-powered meal-planning and diet-tracking application on **AWS PartyRock** using LLMs, supporting personalized plans for **50+ test users** during evaluation and demonstrating applied GenAI product development.

CERTIFICATIONS & BADGES

- Salesforce Certified Agentforce Specialist — *Dec 2025*
- AICTE Virtual Internship - Generative AI (curriculum by Google Cloud) — *Apr 2025*
- SAP ERP Badges — *Apr 2025*
- AWS Educate: Introduction to Generative AI - Training Badge — *Mar 2026*
- Google Cloud Engineering Certificate — *Apr 2026*